# CSE370 Lab Assignment - 03

**Name: MD. Shafiur Rahman**

**Student ID: 21201613**

**Section: 04**

## MySQL commands:

- `mysql -u root -p`
- `""`
- `show databases;`
- `create database` **`Bank;`**
- `use` **`Bank;`**

Used a database called **Bank**.

Data for the **Bank** database have been taken from here : https://docs.google.com/document/d/1dCRpz5RGNrUsB8uJwFgBPCPov8d1FyewIvb6iFaPyQ8/

## Task-1:

Find the ***name*** and ***loan number*** of all customers having a loan at the Downtown branch. [2]

- select customer.**customer_name**, loan.**loan_number**

  from customer, loan, borrower

  where customer.**customer_id**=borrower.**customer_id** and borrower.**loan_number** = loan.**loan_number**

  and loan.**branch_name** = 'Downtown';

```
MariaDB [bank]> select customer.customer_name, loan.loan_number
    -> from customer, loan, borrower
    -> where customer.customer_id=borrower.customer_id and borrower.loan_number = loan.loan_number
    -> and loan.branch_name = 'Downtown';
+---------------+-------------+
| customer_name | loan_number |
+---------------+-------------+
| Johnson       | L-14        |
| Jones         | L-17        |
| Williams      | L-17        |
+---------------+-------------+
3 rows in set (0.022 sec)
```

## Task-2:

Find all the possible pairs of customers who are from the same ***city***. show in the format Customer1, Customer2, City. [2]

- select l1.**customer_name** as **Customer1**, l2.**customer_name** as **Customer2**, l1.**customer_city** as **city**

  from customer l1, customer l2 where

  l1.**customer_city** = l2.**customer_city** and l1.**customer_name**!= l2.**customer_name**

  group by city;

```
MariaDB [bank]> select l1.customer_name as Customer1, l2.customer_name as Customer2, l1.customer_city as city
    -> from customer l1, customer l2 where
    -> l1.customer_city = l2.customer_city and l1.customer_name!= l2.customer_name
    -> group by city;
+-----------+-----------+------------+
| Customer1 | Customer2 | city       |
+-----------+-----------+------------+
| Hayes     | Jones     | Harrison   |
| Adams     | Lindsay   | Pittsfield |
| Curry     | Smith     | Rye        |
| Green     | Turner    | Stamford   |
+-----------+-----------+------------+
4 rows in set (0.001 sec)
```

## Task-3:

If the bank gives out 4% interest to all accounts, show the total interest across each ***branch***. Print ***Branch_name***, ***Total_Interest*** [1]

- `select account.`**`branch_name`**`,`
  `sum(account.`**`balance`**`*(4/100)) as` **`Total_Interest`** `from`
  `account group by` **`branch_name`**`;`

```
MariaDB [bank]> select branch_name, sum(account.balance*(4/100)) as Total_Interest from
    -> account group by branch_name;
+-------------+----------------+
| branch_name | Total_Interest |
+-------------+----------------+
| Brighton    |        66.0000 |
| Downtown    |        20.0000 |
| Mianus      |        28.0000 |
| Perryridge  |        16.0000 |
| Redwood     |        28.0000 |
| Round Hill  |        14.0000 |
+-------------+----------------+
6 rows in set (0.001 sec)
```

## Task-4:

Find account numbers with the highest balances for each ***city*** in the database [1]

- select account.**account_number**, max(account.**balance**) as **balance**, branch.**branch_city**

  from account, branch where account.**branch_name** = branch.**branch_name**

  group by branch.**branch_city**;

```
MariaDB [bank]> select account.account_number, max(account.balance) as balance, branch.branch_city
    -> from account, branch where account.branch_name = branch.branch_name
    -> group by branch.branch_city;
+----------------+---------+-------------+
| account_number | balance | branch_city |
+----------------+---------+-------------+
| A-101          |     900 | Brooklyn    |
| A-102          |     700 | Horseneck   |
| A-222          |     700 | Palo Alto   |
+----------------+---------+-------------+
3 rows in set (0.001 sec)
```

## Task-5:

Show the ***loan number, loan amount, and name*** of customers who have the top 5 highest ***loan amounts***. The data should be sorted by increasing amounts, then decreasing loan numbers in case of the same loan amount. [Hint for top 5 check the "limit" keyword in mysql] [2]

- select loan.**loan_number**, loan.**amount**, customer.**customer_name** from customer, loan, borrower

  where customer.**customer_id**=borrower.**customer_id** and borrower.**loan_number** = loan.**loan_number**

order by **amount** desc limit 5, **loan_number** asc;

```
MariaDB [bank]> select loan.loan_number, loan.amount, customer.customer_name from customer, loan, borrower
    -> where customer.customer_id=borrower.customer_id and borrower.loan_number = loan.loan_number
    -> order by amount desc, loan_number asc limit 5;
+-------------+--------+---------------+
| loan_number | amount | customer_name |
+-------------+--------+---------------+
| L-23        |   2000 | Smith         |
| L-14        |   1500 | Johnson       |
| L-15        |   1500 | Hayes         |
| L-16        |   1300 | Adams         |
| L-17        |   1000 | Jones         |
+-------------+--------+---------------+
5 rows in set (0.001 sec)
```

## Task-6:

Find the names of customers with an ***account*** and also a ***loan*** at the Perryridge ***branch***. [2]

- select customer.**customer_name**, account.**account_number** from customer, account, loan, borrower, depositor

  where customer.**customer_id** = depositor.**customer_id** and depositor.**account_number** = account.**account_number**

  and depositor.**customer_id** = borrower.**customer_id** and borrower.**loan_number** = loan.**loan_number**

  and loan.**branch_name** = 'Perryridge'

```
MariaDB [bank]> select customer.customer_name, account.account_number from customer, account, loan, borrower, depositor
    -> where customer.customer_id = depositor.customer_id and depositor.account_number = account.account_number
    -> and depositor.customer_id = borrower.customer_id and borrower.loan_number = loan.loan_number
    -> and loan.branch_name = 'Perryridge';
+---------------+----------------+
| customer_name | account_number |
+---------------+----------------+
| Hayes         | A-102          |
+---------------+----------------+
1 row in set (0.001 sec)
```

## Task-7:

Find the total ***loan amount*** of all customers having at least 2 loans from the bank. Show in format ***customer name, total_loan***. [2]

- select customer.**customer_name**, count(*) as **number_of_loans**, sum(loan.**amount**) as **total_loan** from customer, loan, borrower

  where customer.**customer_id**=borrower.**customer_id** and borrower.**loan_number** = loan.**loan_number**

  group by customer.**customer_id** having count(customer.**customer_id**) = 2;

```
MariaDB [bank]> select customer.customer_name, count(*) as number_of_loans, sum(loan.amount) as total_loan
    -> from customer, loan, borrower
    -> where customer.customer_id=borrower.customer_id and borrower.loan_number = loan.loan_number
    -> group by customer.customer_id having count(customer.customer_id) = 2;
+---------------+-----------------+------------+
| customer_name | number_of_loans | total_loan |
+---------------+-----------------+------------+
| Smith         |               2 |       2900 |
+---------------+-----------------+------------+
1 row in set (0.001 sec)
```